

Scalability & Future Plan

Date	29 September 2026
Team ID	6ab4ce760fc666a751b52130
Project Name	AI StudyBuddy
Maximum Marks	1 Mark

Current System Limitations

S.No	Limitation	Impact	Priority
1	No frontend UI — backend API only, tested via Postman	Students cannot interact with the platform visually; limits user adoption	High
2	Basic PDF text extraction (plain text only, no OCR for images/scanned PDFs)	Complex PDFs with embedded images or scanned pages may not extract fully	Medium
3	Single-server local deployment — not hosted on cloud for public access	Only accessible on localhost; cannot serve remote users	High

Scalability Plan

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Single-user local testing via Postman	Deploy on cloud (AWS/Render/Vercel) with load balancing. Express.js supports clustering for multi-core utilization.
2	Data Storage	MongoDB Atlas cloud database (free tier)	Upgrade to MongoDB Atlas dedicated cluster with auto-scaling, sharding for large datasets, and automated backups.
3	Performance	AI generation takes 3-8 seconds per request (Gemini API latency)	Implement response caching for repeated queries. Use Redis for session management. Queue AI requests with Bull/BullMQ for high concurrency.
4	Security	JWT with bcrypt, environment variables for secrets	Add rate limiting (express-rate-limit), input sanitization (helmet, express-validator), HTTPS enforcement, and API key rotation.

Future Roadmap

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	Frontend UI using React.js or Next.js — visual interface for registration, material upload, and viewing AI-generated content	3 months	Dramatically improves user experience and adoption. Students interact visually instead of via Postman.
Phase 3	AI Chatbot for interactive doubt solving — real-time Q&A with Gemini about uploaded materials. Progress tracking dashboard — track study completion and quiz scores over time.	6 months	Adds interactive learning capabilities. Progress tracking motivates students and identifies weak areas.
Phase 4	Mobile app (React Native). Multi-language support for AI content generation. Collaborative study groups — share materials and AI content with peers. LMS integration (Moodle, Google Classroom).	12 months	Expands platform reach to mobile users. Multi-language support serves diverse student populations. LMS integration brings AI StudyBuddy into existing educational workflows.